MART-NILM: A Compact Multi-Appliance Relational Temporal Network for Joint Power-State Estimation in Non-Intrusive Load Monitoring

Raymond Siou Hin Tie, Wen-Shan Tan, Ze Yang Ding, Mirza Rayana Sanzana, and Haval Kamil

***Abstract*-Appliance-level visibility supports residential demand-side management, but aggregate meters do not directly reveal individual consumption or operating states. Non-intrusive load monitoring (NILM) estimates these quantities without dedicated sensors. Deep multi-appliance NILM uses one neural model to jointly estimate appliance power and multi-label ON-OFF states from aggregate measurements, but sparse activations under limited background conditions and unequal power-state loss scales complicate training. This paper proposes the Multi-Appliance Relational Temporal Network for NILM (MART-NILM), a compact estimator with a shared temporal convolutional encoder, appliance-specific decoders, and relational attention. Focal-event background-stratified mixing varies background demand around active appliance traces, while bounded appliance-wise balancing calibrates each state loss relative to its corresponding power loss. Both operate only during training. On two public residential datasets with disjoint training, validation, and test houses, MART-NILM attains appliance-average mean absolute errors (MAEs) of 7.3 and 10.9 W and macro-F1 scores of 0.88 and 0.72, outperforming three reproduced baselines. With 1.373 million parameters, it also records lower observed training and inference times under the evaluated protocol. These results support compact joint power-state estimation for residential load monitoring.**

***Index Terms*-non-intrusive load monitoring, energy disaggregation, multi-task learning, power and state prediction, household appliances**

I. Introduction

Effective residential demand-side management requires visibility into the electricity consumption and operating status of individual appliances. Power estimates quantify appliance demand, whereas state estimates expose activation timing; aggregate metering reveals neither directly. Appliance-level feedback can support energy conservation [1], but dedicated submetering requires installing and maintaining additional sensors. Non-intrusive load monitoring (NILM) commonly estimates both appliance power consumption and operating status from aggregate measurements, avoiding the need for dedicated appliance-level submetering [2]. Joint estimation of these outputs is important because a low average power error can still mask missed operating periods, especially for sparsely active appliances.

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Classical NILM identifies appliance signatures and infers appliance states using finite-state or factorial hidden Markov models [3], [4]. Deep approaches instead learn representations directly from aggregate windows [5], reducing the need to specify signatures and state-transition models manually. Regardless of the method used, the aggregate power measured at the main meter includes demand from loads outside the monitored target set. For N target appliances, the measured power is

$$x_t = \sum_{i=1}^{N} P_{t,i} + b_t + \epsilon_t, \tag{1}$$

where $P_{t,i}$ is target power, b_t is background demand from unmonitored loads, and ϵ_t represents measurement and timing discrepancies. Across houses, the background demand b_t may exceed the power of a low-power target appliance and may follow a different distribution. Consequently, a model trained in one house may encounter unfamiliar background demand in another, even when the target appliance is unchanged. Such cross-house variation makes NILM challenging because the target appliance must be distinguished from unfamiliar background demand.

Against this background, deep NILM commonly considers two complementary tasks, power estimation and ON-OFF state detection, and is formulated as either single-appliance or multi-appliance estimation. Single-appliance networks train one model for each appliance and may yield better appliance-level disaggregation accuracy in some settings [6], [7], but repeating a model for every load duplicates aggregate feature extraction and storage, which is undesirable for smart-meter or home-gateway deployment. Recent deep NILM work has therefore increasingly studied multi-appliance estimation, in which one model jointly estimates the power traces and ON-state probabilities of a fixed set of N target appliances from the same aggregate window [8]-[11].

Joint multi-appliance and multi-label estimation is more challenging than running N independent predictors. Concurrent activations provide useful context, but superposed transitions and background demand can mask weak targets [12]. Therefore, a multi-appliance model must distinguish appliance-specific signatures from overlapping demand, while exploiting relationships among concurrently active appliances and producing consistent power and state predictions.

Two further challenges arise when training a joint multi-appliance power-state model. First, sparse appliance activations in the available training data are observed under only

a subset of background-demand conditions. The model may therefore use background-demand correlations as shortcuts instead of learning the appliance signature; such shortcuts can fail when those correlations change across households [12]. Second, appliance power ranges and ON-state frequencies differ substantially. Existing joint objectives use a shared regression-classification balance, either as a fixed coefficient or a global loss-ratio weight [11], [13]. Such a shared balance cannot represent appliance-specific scales; when derived from joint losses, a large power error for one appliance also changes the relative state weight of the others. These observations motivate controlling background-demand conditions and balancing the power and state objectives separately for each appliance.

To address these issues, we propose the Multi-Appliance Relational Temporal Network for NILM (MART-NILM), a relational temporal model for joint power and ON-OFF state estimation under varying background demand.

The main contributions are threefold:

1) A compact multi-appliance architecture, MART-NILM, is designed for joint power and state estimation. A shared convolutional temporal encoder avoids duplicating the temporal backbone for each appliance, while appliance-specific decoders and cross-appliance attention retain appliance specialization and relational context. The model produces five power-state output pairs in a single forward pass with 1.373 million parameters.
2) Two training-only methods address two distinct training challenges. First, focal-event background-stratified mixing (hereafter focal-event mixing) addresses sparse appliance activations and limited background-demand coverage by pairing selected ON events with diverse recorded background demand. Second, bounded appliance-wise balancing addresses appliance-specific power-state loss imbalance by separately calibrating the power and state objectives for each appliance. Neither method adds inference-time computation.
3) A cross-house evaluation is conducted using aggregate power recordings from two public residential datasets. Baseline comparisons are repeated over three random seeds. Architecture and training-strategy ablations, together with analyses stratified by background demand, assess power-estimation and state-detection performance, computational cost, and residual failure modes.

II. Related Work

Viewed from a multi-task learning (MTL) perspective [14], joint multi-appliance power-state estimation combines two interacting design levels. Within each appliance, power regression and state classification must be coordinated; across appliances, feature sharing must reduce duplicated computation without suppressing appliance-specific information. This nested setting has been described as "multi-task-multi-appliance" [8] and "multi-appliance-task" [11], [15]. Prior NILM architectures are commonly distinguished by their temporal backbones, including convolutional neural networks (CNNs) [8], [15], recurrent neural networks (RNNs) [16], temporal convolutional networks (TCNs) [9], [17], and Transformers [11]. For joint multi-appliance power-state estimation, however, backbone choice addresses only temporal feature extraction. The central design questions are how the appliance head couples power regression with state classification and how the model body shares, specializes, and exchanges representations across appliances. Training strategies must additionally address sparse appliance events, limited background-demand coverage, and unequal power-state loss scales. The following review therefore considers appliance-head design, cross-appliance body design, and training strategies for joint power-state estimation.

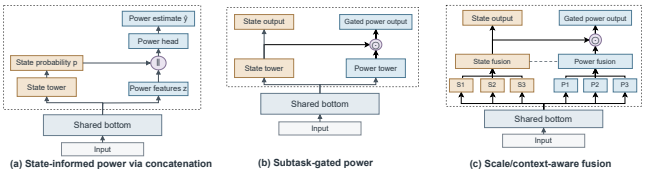


Fig. 1. Schematic within-appliance power-state heads: (a) state-informed concatenation (Murray *et al.* [16]); (b) output gating (SGN [18]); and (c) multi-scale fusion (SCANet [19]).

A. Appliance-Head Design

Existing joint power-state NILM studies have mainly examined how state information should influence power estimation within an individual target model. Figure 1 organizes representative appliance-head designs according to where this coupling is introduced.

Murray *et al.* [16] concatenate predicted states with power features (Fig. 1(a)), but the regressor remains free to produce nonzero power for an OFF prediction. The subtask-gated network (SGN) [18] instead gates regression outputs with ON-state probabilities (Fig. 1(b)), suppressing inactive power while making reconstruction sensitive to missed activations. The scale- and context-aware network (SCANet) [19] adds parallel dilated branches and self-attention (Fig. 1(c)), increasing processing within each target model. These designs address within-appliance power-state coupling, but their original single-appliance formulations require separate models for different targets, repeating aggregate feature extraction and providing no explicit exchange among appliance-specific representations.

B. Cross-Appliance Body Design

Cross-appliance body design determines which representations remain shared, which are decoded separately, and where appliance-specific streams exchange information. Multi-Appliance-Task NILM (MATNilm) [15] reduces this duplication through shared encoding and private appliance decoders (Fig. 2(a)). Its temporal and appliance-wise attention captures both dependencies, but repeats temporal attention and feed-forward processing in the private branches. Sun *et al.* [11] use the same sharing pattern with Transformer blocks in shared and appliance-specific layers (Fig. 2(a)). Downsampling reduces shared attention cost, but temporal attention is still

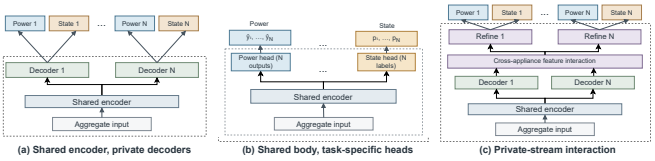


Fig. 2. Schematic multi-appliance body mechanisms: (a) shared encoder/private decoders (MATNilm [15] and Sun *et al.* [11]); (b) shared body/task heads (UNet-NILM [8]); and (c) private-stream interaction (MATNilm [15] and MART-NILM).

repeated for individual appliances. Both support joint multi-appliance power and state estimation; their private sequence-processing blocks nevertheless add computation beyond the shared encoder.

UNet-NILM [8] follows the shared-body/task-head pattern (Fig. 2(b)), predicting multiple appliance powers and states from common features. It avoids private temporal blocks, but leaves appliance differentiation to shared features and output heads, without dedicated appliance-specific decoder streams or explicit exchange between such streams.

Cross-appliance interaction (Fig. 2(c)) complements these designs by exchanging context among specialized representations. MATNilm already uses appliance-wise attention for this purpose [15], while message passing provides a related mechanism in dense prediction [21]. These designs motivate examining how appliance-specific decoding and relation attention can be retained while reducing repeated temporal processing.

C. Training Strategies for Joint Power-State Estimation

Training a joint multi-appliance power-state model also requires sufficient exposure to appliance activations under different background-demand conditions. Neural NILM constructs synthetic aggregates to increase training diversity [5], SCANet augments ON-state examples [19], and aggregate recomposition varies background demand while preserving target labels [12]. These approaches address limited training coverage, but input diversity alone does not ensure that sparse ON events are sampled across a controlled range of recorded background demand. This distinction matters when event-background correlations change across houses.

A separate issue is the unequal scale of power and state losses. Gradient normalization (GradNorm) balances task updates in generic MTL [22], whereas SAMNet adapts a power-state loss ratio for joint NILM [13]. In a multi-appliance setting, a balance derived from aggregate losses may be driven by appliances with larger regression losses and may not reflect the power-state loss ratio of each appliance. This motivates appliance-wise balancing with bounds that limit extreme weights.

Together, these studies motivate the architectural and training choices examined in this paper. A shared temporal encoder, appliance-specific decoders, and relation attention address representation design, while focal-event mixing and bounded appliance-wise balancing address background-demand coverage and appliance-specific power-state loss scales, respectively.

III. Methodology

A. Problem Formulation

Following Equation (1), let $\mathbf{x} = (x_1, \ldots, x_T)$ denote an aggregate window of length T. The problem is to jointly estimate the power traces and ON-state probabilities of N target appliances:

$$\left(\hat{\mathbf{Y}}, \hat{\mathbf{S}}\right) = f_{\boldsymbol{\theta}}(\mathbf{x}), \qquad \hat{\mathbf{Y}} \in \mathbb{R}^{T\times N}, \quad \hat{\mathbf{S}} \in [0,1]^{T\times N} \tag{2}$$

Here $\boldsymbol{\theta}$ denotes the model parameters, and $\hat{\mathbf{Y}}$ and $\hat{\mathbf{S}}$ collect the estimated power traces and ON-state probabilities, respectively. To make the multi-appliance and multi-label output structure explicit, the time-t output slice of (2) is

$$f_{\boldsymbol{\theta},t}(\mathbf{x}) = \underbrace{\left(\hat{y}_{t,1}, \hat{s}_{t,1}, \ldots, \hat{y}_{t,N}, \hat{s}_{t,N}\right)}_{N \text{ appliance power-state pairs}}, \quad t = 1, \ldots, T. \tag{3}$$

Thus, for appliance i at time t, $\hat{y}_{t,i}$ is the estimated standardized power and $\hat{s}_{t,i} \in [0,1]$ is its ON-state probability. Equations (2) and (3) define the objective: from one aggregate window, jointly recover the power trace and ON-state probability of every target appliance at every time step. Power estimation is a regression task, whereas state detection is multi-label classification because multiple appliances may be ON simultaneously. Binary ON-OFF labels summarize activity without resolving internal operating modes.

B. Model Architecture

The implementation uses instance-batch normalization (IBN), batch normalization (BN), the rectified linear unit (ReLU) activation, and fully connected (FC) output layers.

Figure 3 presents MART-NILM, combining the private-stream interaction body in Fig. 2(c) with the state-gated heads in Fig. 1(b). The components are described below in processing order, using the same names as the architecture-ablation variants in Table V.

Multi-scale CNN: Parallel convolutional branches [24] extract local features at complementary time scales from the standardized aggregate $\bar{\mathbf{x}}$:

$$\mathbf{Z} = \mathcal{F}([\mathcal{B}_3(\bar{\mathbf{x}}); \mathcal{B}_5(\bar{\mathbf{x}}); \mathcal{B}_9(\bar{\mathbf{x}})]) + \mathcal{S}(\bar{\mathbf{x}}). \tag{4}$$

Here $\mathcal{B}_k$ is a kernel-k convolution followed by IBN [25] and ReLU; $[\cdot;\cdot]$ denotes channel concatenation. The 1×1 fusion $\mathcal{F}$ combines the branches, while the projected shortcut $\mathcal{S}$ retains a direct input path. IBN applies instance normalization (IN) and BN to separate channel halves.

Feature Encoder: Two convolution-IBN-ReLU stages combine the local features into a shared representation $\mathbf{H}^{(0)} \in \mathbb{R}^{128\times T}$. Feature extraction is performed once for all appliances.

Shared Temporal Encoder: Residual dilated TCN blocks [26] extend temporal context:

$$\mathbf{H}^{(\ell+1)} = \mathbf{H}^{(\ell)} + \mathcal{T}_{2^\ell}(\mathbf{H}^{(\ell)}), \qquad \ell \in \{0,1,2,3,4\}. \tag{5}$$

Here $\mathcal{T}_d$ comprises a dilation-d convolution, BN, ReLU, and dropout. Increasing dilation expands context, while residual connections and symmetric padding retain the incoming features and temporal alignment. The output is $\mathbf{H} = \mathbf{H}^{(5)}$.

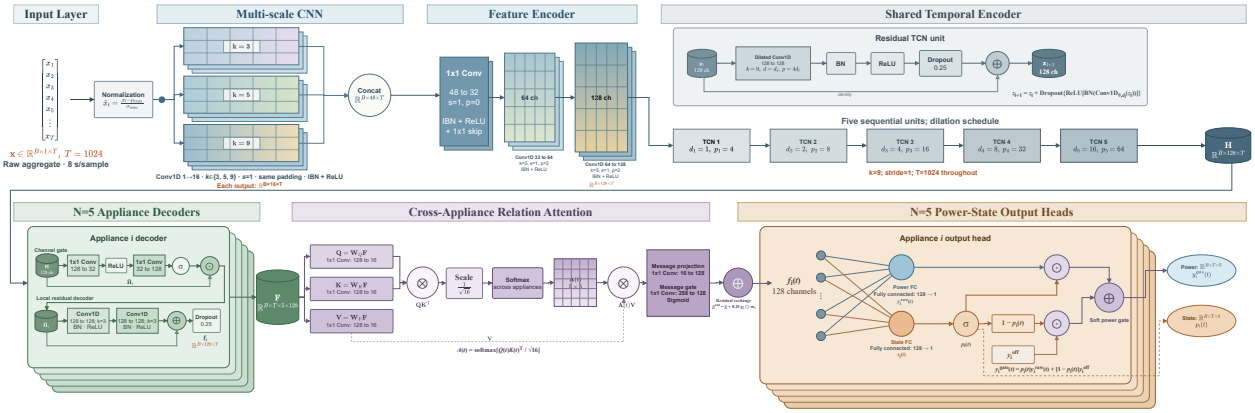


Fig. 3. Model architecture of MART-NILM.

Appliance Decoders: Each decoder applies a channel gate, inspired by channel recalibration [27], and a local residual transformation to obtain an appliance-specific representation $\mathbf{F}_i$. These private streams adapt the shared temporal features to individual load characteristics.

Cross-Appliance Relation Attention: The relation module draws on the cross-task information-sharing principle of the Multi-Tasks Guided Prediction-and-Distillation Network (PAD-Net) [21] to exchange context among appliance-specific feature streams. Scaled dot-product attention [28] is evaluated across appliances at each time t:

$$\mathbf{A}_t = \mathrm{softmax}\left(\frac{\mathbf{Q}_t\mathbf{K}_t^{\mathsf{T}}}{\sqrt{d}}\right), \qquad \mathbf{M}_t = \mathbf{A}_t\mathbf{V}_t. \tag{6}$$

Here $\mathbf{Q}_t$, $\mathbf{K}_t$, and $\mathbf{V}_t$ are learned projections of the stacked appliance representations, and d is their projection width. The attended context is projected back to the feature dimension and added to each private stream through a gated residual connection.

Power-State Output Heads: For each appliance, two time-shared FC heads predict raw standardized power $\hat{y}_i^{\mathrm{raw}}(t)$ and a state logit $a_i(t)$. Following the state-gating principle of SGN [18], the ON-state probability $\hat{s}_i(t) = \mathrm{sigmoid}(a_i(t))$ softly gates the power prediction:

$$\hat{y}_i(t) = \hat{s}_i(t)\hat{y}_i^{\mathrm{raw}}(t) + [1 - \hat{s}_i(t)]y_i^{(0)}, \tag{7}$$

where $y_i^{(0)} = -\mu_i/\sigma_i$ is the standardized zero-power anchor. Hence, a low ON-state probability smoothly moves the estimate towards zero power. Predictions are converted back to watts using the appliance training mean μ_i and standard deviation σ_i, and negative values are clipped to zero. The gate couples the state and power predictions while retaining a separate waveform regressor.

C. Loss Function

The model is trained jointly over all appliances, as in multi-task NILM [8], [11], [15]. Let $\mathcal{L}_{\mathrm{power},i}$ and $\mathcal{L}_{\mathrm{state},i}$ denote the power and state losses for appliance i, respectively. The joint objective is

$$\mathcal{L} = \sum_{i=1}^{N} \left(\mathcal{L}_{\mathrm{power},i} + \lambda r_i \mathcal{L}_{\mathrm{state},i}\right), \qquad \lambda = 0.8. \tag{8}$$

Here r_i is the appliance-wise balancing factor defined in Section III-D.

Power loss: The power objective combines waveform fidelity and switching fidelity:

$$\mathcal{L}_{\mathrm{power},i} = \underbrace{\mathcal{L}_{\mathrm{wave},i}}_{\text{power-level fidelity}} + \underbrace{0.15\,\mathcal{L}_{\mathrm{trans},i}}_{\text{transition fidelity}}. \tag{9}$$

The waveform and transition terms both use mean squared error (MSE), applied to power values and temporal differences, respectively. The expressions below are written for one window; in implementation, both terms are averaged over the batch and time dimensions. Let $y_i(t)$ and $\hat{y}_i(t)$ denote standardized target and predicted power, and let $z_i(t) \in \{0, 1\}$ denote the ON-OFF label. For a temporal index set $\mathcal{S}$, define

$$\mathrm{MSE}_{\mathcal{S}}(u, v) = \frac{1}{\max(|\mathcal{S}|, 1)} \sum_{t\in\mathcal{S}} \big(u(t) - v(t)\big)^2. \tag{10}$$

The waveform term evaluates power errors over all samples $\mathcal{T}_0 = \{1, \ldots, T\}$, ON-state samples $\mathcal{T}_i^{\mathrm{on}} = \{t \in \mathcal{T}_0 : z_i(t) = 1\}$, and OFF-state samples $\mathcal{T}_i^{\mathrm{off}} = \{t \in \mathcal{T}_0 : z_i(t) = 0\}$:

$$\begin{aligned}\mathcal{L}_{\mathrm{wave},i} = {} & \mathrm{MSE}_{\mathcal{T}_0}(y_i, \hat{y}_i) + \mathrm{MSE}_{\mathcal{T}_i^{\mathrm{on}}}(y_i, \hat{y}_i) \\ & + 0.5\ \mathrm{MSE}_{\mathcal{T}_i^{\mathrm{off}}}(y_i, \hat{y}_i).\end{aligned} \tag{11}$$

The first term measures global waveform fidelity; the ON-state term prevents sparse activations from being dominated by OFF samples, and the lower-weight OFF-state term constrains inactive predictions.

To additionally constrain changes in power, let $\Delta y_i(t) = y_i(t) - y_i(t-1)$ and define $\Delta\hat{y}_i(t)$ analogously. The transition term compares these differences whenever the appliance is ON at either adjacent sample, over $\mathcal{T}_i^{\Delta} = \{t \in \{2, \ldots, T\} : z_i(t) = 1 \text{ or } z_i(t-1) = 1\}$:

$$\mathcal{L}_{\mathrm{trans},i} = \mathrm{MSE}_{\mathcal{T}_i^{\Delta}}(\Delta y_i, \Delta\hat{y}_i). \tag{12}$$

State loss: The state objective combines weighted binary cross-entropy (BCE) with explicit false-positive control:

$$\mathcal{L}_{\mathrm{state},i} = \underbrace{\mathcal{L}_{\mathrm{BCE},i}}_{\text{class-balanced state detection}} + \underbrace{\mathcal{L}_{\mathrm{fp},i}}_{\text{false-positive control}}. \tag{13}$$

For a mini-batch of B windows, let $\hat{s}_{b,t,i} \in (0, 1)$ be the predicted ON-state probability and $z_{b,t,i} \in \{0, 1\}$ the target label.

The class-balanced BCE term uses $w_i = \min\{(1-\pi_i)/\pi_i, 12\}$, determined by the training ON-state proportion π_i:

$$\mathcal{L}_{\mathrm{BCE},i} = -\frac{1}{BT}\sum_{b,t}\big[w_i z_{b,t,i}\log \hat{s}_{b,t,i} + (1-z_{b,t,i})\log(1-\hat{s}_{b,t,i})\big]. \tag{14}$$

The false-positive term penalizes confident ON-state predictions on OFF samples:

$$\mathcal{L}_{\mathrm{fp},i} = \frac{\sum_{b,t}(1-z_{b,t,i})\hat{s}^2_{b,t,i}}{\max\left(\sum_{b,t}(1-z_{b,t,i}), 1\right)}. \tag{15}$$

D. Bounded Appliance-Wise Balancing

To account for the different numerical scales of regression and classification losses, a bounded loss-ratio weighting is used. It is motivated by loss-ratio balancing in SAMNet [13] and differs from gradient-norm balancing in GradNorm [22]. The global reference ratio g bounds the appliance-specific ratio r_i:

$$g = \frac{\sum_j \mathcal{L}_{\mathrm{power},j}}{\sum_j \mathcal{L}_{\mathrm{state},j}}, \qquad r_i = \min\left\{3g, \max\left\{\frac{g}{3}, \frac{\mathcal{L}_{\mathrm{power},i}}{\mathcal{L}_{\mathrm{state},i}}\right\}\right\}. \tag{16}$$

At each mini-batch update, g and r_i are evaluated from the current losses and held fixed during back-propagation. The three-fold bounds limit each local ratio to $[g/3, 3g]$, preventing extreme weights from dominating optimization.

E. Focal-Event Background-Stratified Mixing

Training uses recorded and recomposed aggregate windows with equal probability. Recorded windows preserve the observed joint demand. These windows extend synthetic aggregate construction in Neural NILM [5] by pairing appliance events with recorded background demand from other training windows [12], reducing reliance on house-specific event-background correlations. For a recorded sequence, the background demand is estimated by the nonnegative residual

$$R_t = \left[x_t - \sum_{i=1}^{N} P_{t,i}\right]_+. \tag{17}$$

Here $[u]_+ = \max(u, 0)$. Unlike b_t in Equation (1), R_t can also include measurement and timing discrepancies. For each recomposed window, the residual and appliance traces are drawn from independently selected training windows:

$$\tilde{x}_t = R_t^{(k_0)} + \sum_{i=1}^{N} P_{t,i}^{(k_i)}, \qquad \tilde{P}_{t,i} = P_{t,i}^{(k_i)}, \quad \tilde{z}_{t,i} = z_{t,i}^{(k_i)}, \tag{18}$$

where $k_0, k_1, \ldots, k_N$ denote independently sampled window indices. In Step 1 of Fig. 4, one focal appliance $i^\star$ is selected uniformly, and its trace is drawn from a window containing at least one ON sample; the remaining appliance traces are sampled independently. In Step 2, a non-empty bin of median residual power is selected uniformly from $[0, 100)$, $[100, 200)$, $[200, 400)$, $[400, 800)$, and $[800, \infty)$ W, and a background window is sampled uniformly within that bin. The selected traces and background are then added according to Equation (18), as illustrated in the lower panel of Fig. 4. The power and state labels of every selected appliance trace are retained. Recomposition is used only during training; validation and test use recorded aggregates and aligned labels.

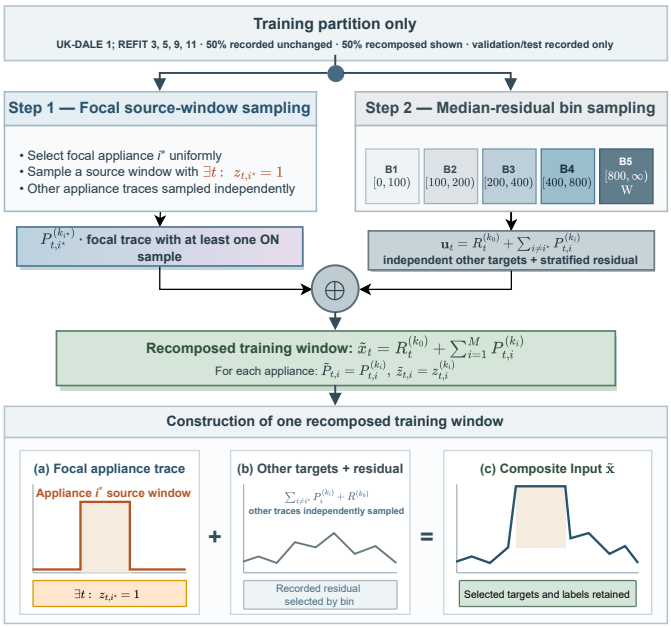


Fig. 4. Focal-event background-stratified mixing. A focal ON-event trace, independently sampled target traces, and a residual window selected by median-power bin form a recomposed training window.

IV. Experimental Evaluation

A. Dataset

Two public residential datasets are used: the UK Domestic Appliance-Level Electricity (UK-DALE) dataset [29] and the REFIT electrical-load dataset [30]. The target appliances are the kettle, fridge, dishwasher, washer, and microwave, and all signals are resampled at 8 s.

The experiment uses a whole-house split: UK-DALE house 1 and REFIT houses 3, 5, 9, and 11 for training; UK-DALE house 5 and REFIT house 2 for validation; and UK-DALE house 2 and REFIT house 20 for testing. The resulting training-validation-test ratio is 5:2:2 by house count. Because each house contributes one continuous 5-week block, this also corresponds to 55.6%, 22.2%, and 22.2% of the nominal recording duration, respectively. The held-out blocks are centred within the previously established eight-week recording intervals and are selected without reference to appliance events or model performance. Normalization statistics are estimated from the training houses only.

B. Data Preprocessing

After 8 s resampling, no additional temporal alignment is applied; residual timing discrepancies remain a measurement

TABLE I
ON-OFF LABEL PARAMETERS. τ^P, G, AND D DENOTE THE ON-POWER THRESHOLD, MAXIMUM OFF-GAP DURATION, AND MINIMUM ON DURATION, RESPECTIVELY. THE FINAL COLUMN LISTS UK-DALE / REFIT VALUES.

Appliance	τ^P (W)	G (s)	D (s)
Kettle	200	0	12 / 12
Fridge†	50	12	60 / 60
Dishwasher†	50	1800	960 / 960
Washer†	20	960	960 / 960
Microwave	200	30	12 / 6

† For each sample, a five-sample local-median filter sets its label-generation power to zero when at least 60% of its four surrounding values are below 50 W and the sample exceeds both three times their median and 100 W. Endpoint windows use edge replication.

limitation. Rows missing any required aggregate, appliance-power, or state-label value are excluded, and windows do not cross timestamp discontinuities.

Let n index windows and t index samples within a window. The aggregate mean and standard deviation, μ_x and σ_x, and the corresponding appliance-i statistics, μ_i and σ_i, are estimated from the training houses. The standardized input power, $\bar{x}_{n,t}$, and target power, $y_{n,t,i}$, are defined as

$$\bar{x}_{n,t} = \frac{x_{n,t} - \mu_x}{\sigma_x}, \qquad y_{n,t,i} = \frac{P_{n,t,i} - \mu_i}{\sigma_i}. \tag{19}$$

State labels are precomputed before windowing. For appliances marked in Table I, isolated spikes are first removed on a label-only copy of the resampled power trace. Each sample is then labeled ON when this trace reaches its appliance-specific threshold τ_i^P. OFF gaps of at most G_i are merged; subsequently, complete ON runs shorter than D_i are removed. Runs touching a sequence boundary are retained because their full duration is unobserved. Table I specifies the physical-time values of these parameters, which are converted to 8 s samples by half-up rounding before labeling. UK-DALE house 5 uses a 30 s fridge gap.

C. Experimental Setup

All experiments were conducted on a single NVIDIA GeForce RTX 4090 graphics processing unit (GPU; 24 GB) using PyTorch 2.6.0 with Compute Unified Device Architecture (CUDA) 12.4. MART-NILM is compared with UNet-NILM [8], the convolutional MATNilm variant (MAT-Conv) [15], and the multi-appliance-task network of Sun *et al.* [11]. Each baseline retains the architecture, objective, and training hyperparameters of its source paper, while all models use the common cross-house data protocol. Table II lists their settings.

MART-NILM uses weight decay 10^{-4} and unit-norm gradient clipping. Its learning rate is reduced by a factor of 0.5 after 12 epochs without validation improvement, to a minimum of 10^{-5}. Training is limited to 150 epochs; early stopping is disabled before epoch 100 and thereafter uses a patience of 55 epochs. At 8 s per sample, each 1024-sample input-output window spans 136.5 min of household measurements.

For MART-NILM, checkpoint selection uses only the validation houses. Mean absolute error (MAE) in standardized power coordinates and average precision (AP) are computed from the validation outputs, and the retained checkpoint minimizes

$$S_{\text{val}} = \text{MAE}_{\text{val}}^{\text{norm}} + 1 - \text{AP}_{\text{val}}^{\text{macro}}, \tag{20}$$

where $\text{MAE}_{\text{val}}^{\text{norm}}$ is the macro mean absolute error over appliances in standardized power coordinates, and the AP term summarizes the precision-recall curve without selecting a decision threshold. For each model and appliance i, let $\hat{s}_{t,i} \in [0,1]$ denote the predicted ON probability at sample t. After checkpoint selection, an appliance-specific threshold $\gamma_i \in \{0.05, 0.06, \ldots, 0.98\}$ is selected on the validation houses to maximize appliance-level F1 after fixed temporal post-processing is applied. At test time, the decoded state label, $\hat{z}_{t,i}$, is obtained as

$$\hat{z}_{t,i} = \mathbb{I}[\hat{s}_{t,i} \geq \gamma_i], \tag{21}$$

where $\mathbb{I}[\cdot]$ is the indicator function. For the microwave, fixed temporal post-processing removes predicted ON runs shorter than 32 s and merges OFF gaps of at most 24 s. The decoded state is also applied to the power output, $\hat{P}_{t,i}^{\text{eval}} = \hat{z}_{t,i}\hat{P}_{t,i}$. Thresholds and temporal rules are frozen before testing and applied unchanged to both test houses; test labels are not used to select γ_i.

TABLE II
TRAINING WINDOWING AND BATCH SETTINGS. ALL WINDOW LENGTHS AND STRIDES ARE IN SAMPLES.

Model	Input	Output	Stride	Batch	Optimizer	Epochs
UNet-NILM [8]	100	1	1	128	Adam	50
MAT-Conv [15]	864	64	64	64	Adam	200
Sun *et al.* [11]	480	480	240	32	Adam	100
MART-NILM	1024	1024	512	64	Adam	150

D. Evaluation Metrics

Each appliance is evaluated on power and on the ON-OFF state. For an evaluated record of length T_{rec}, P_t is the recorded power in watts and $\hat{P}_t^{\text{eval}}$ is the state-decoded prediction. The label z_t equals 1 when the appliance is recorded ON and 0 when it is OFF. $\hat{z}_t$ is the predicted label.

Metrics are computed separately for the held-out UK-DALE house 2 and REFIT house 20. Within each test house, the average (Avg.) column is the macro average over the five appliances; architecture-ablation summaries give the two houses equal weight. Performance metrics are reported as mean ± standard deviation over three random seeds on the fixed cross-house split, so the standard deviation reflects optimization variability. The evaluation considers complementary power-estimation and state-detection metrics.

1) Power: Power error is reported using MAE, ON-state MAE (ON MAE), and blockwise signal aggregate error (SAE). MAE uses every sample; ON MAE is the same quantity evaluated only at samples with $z_t = 1$, so that reconstruction quality during appliance operation remains visible:

$$\text{MAE} = \frac{1}{T_{\text{rec}}} \sum_{t=1}^{T_{\text{rec}}} \left| P_t - \hat{P}_t^{\text{eval}} \right|. \tag{22}$$

Following MATNilm [15], SAE measures the mean absolute error of average power over non-overlapping 160 min blocks ($L = 1200$ samples):

$$\mathrm{SAE} = \frac{1}{K}\sum_{k=1}^{K}\left|\frac{1}{L}\sum_{t=(k-1)L+1}^{kL}\left(P_t - \hat{P}_t^{\mathrm{eval}}\right)\right|, \qquad K = \left\lfloor \frac{T_{\mathrm{rec}}}{L} \right\rfloor \tag{23}$$

2) On-Off State: The recorded label z_t follows the rules in Table I, and $\hat{z}_t$ is the predicted binary state. For each appliance, TP (true positives) counts samples correctly identified as ON ($z_t = 1, \hat{z}_t = 1$), FP (false positives) counts inactive samples falsely detected as ON ($z_t = 0, \hat{z}_t = 1$), and FN (false negatives) counts active samples missed as OFF ($z_t = 1, \hat{z}_t = 0$):

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \qquad \mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}. \tag{24}$$

$$\mathrm{F1} = \frac{2\cdot\mathrm{Precision}\cdot\mathrm{Recall}}{\mathrm{Precision}+\mathrm{Recall}}. \tag{25}$$

Precision is the fraction of predicted ON samples that are truly ON, and recall is the fraction of actual ON samples detected. F1 is their harmonic mean; true negatives (TN, $z_t = 0, \hat{z}_t = 0$) are excluded to prevent long idle background periods from artificially inflating the score. To evaluate state ranking independently of the calibrated decision threshold, we also report AP from the continuous state probability. Let $\mathcal{P}_n$ and $\mathcal{R}_n$ denote precision and recall after the nth distinct probability threshold, with $\mathcal{R}_0 = 0$. The non-interpolated AP is

$$\mathrm{AP} = \sum_{n}\left(\mathcal{R}_n - \mathcal{R}_{n-1}\right)\mathcal{P}_n. \tag{26}$$

Unlike F1, AP does not use the appliance-specific threshold or the temporal state post-processing fitted on the validation houses.

V. Results and Discussion

A. Joint Power and State Estimation

Table III reports held-out test performance; lower power errors and higher state scores are better. A displayed zero standard deviation indicates a value below the shown precision.

Across the appliance averages, MART-NILM outperforms MAT-Conv, the strongest baseline. Relative to MAT-Conv, MART-NILM reduces MAE by 23.6% on UK-DALE and 14.5% on REFIT. The corresponding reductions in ON MAE are 58.4% and 34.4%, respectively, and those in SAE are 27.0% and 27.8%, respectively. Macro-F1 increases from 0.72 to 0.88 on UK-DALE and from 0.58 to 0.72 on REFIT. Threshold-independent macro-AP similarly increases from 0.812 to 0.926 on UK-DALE and from 0.576 to 0.761 on REFIT. MART-NILM also achieves the highest displayed appliance-level AP for every appliance. The improvements in both F1 and AP show that the state-detection gain cannot be attributed solely to threshold calibration.

The advantage is not uniform across appliances. MAT-Conv remains better for UK-DALE fridge power and for REFIT kettle and microwave MAE, and its REFIT fridge F1 is marginally higher. MART-NILM instead obtains its clearest gains on dishwasher and washer estimation and on the macro averages. These exceptions show that the proposed representation improves the overall power-state balance rather than dominating every appliance-metric pair.

Figure 5 provides a qualitative view of the same comparison. MART-NILM generally follows activation boundaries and multi-stage plateaus while suppressing inactive-period power, although amplitude and timing errors remain.

B. Performance and Model Efficiency

Table IV shows that MART-NILM contains 1.373 million parameters, reductions of 54.6%, 76.5%, and 82.5% relative to MAT-Conv, UNet-NILM, and the network of Sun *et al.*, respectively. Combined with the accuracy results in Table III, these measurements show that the gains do not require increased model capacity. MART-NILM also has the smallest checkpoint and the shortest observed training and end-to-end inference times. Processing both test records requires 1.14 s, compared with 14.67 s for MAT-Conv, 23.50 s for Sun *et al.*, and 80.40 s for UNet-NILM. The benchmark includes data transfer and output reconstruction under each model's native windowing protocol, so it measures the complete prediction pipeline rather than isolated forward-pass latency. MART-NILM benefits from producing long output sequences: its 1024-sample windows advance by 512 samples, requiring substantially fewer windows than sequence-to-point inference while preserving dense temporal predictions.

C. Architecture Ablation Study

Each variant in Table V removes or simplifies one top-level component while preserving the joint five-appliance output structure and compatible feature dimensions.

The components make complementary contributions rather than improving every metric uniformly. Temporal modeling has the largest effect: removing it increases MAE by 1.44 W and lowers macro-F1 from 0.796 to 0.710. Without appliance-specific decoders, ON MAE increases by 22.90 W and macro-F1 falls to 0.783, supporting explicit separation of appliance representations. The feature encoder improves all four aggregate measures. Replacing the multi-scale CNN with a single-scale variant slightly worsens MAE, SAE, and macro-F1 while marginally improving ON MAE, indicating a modest rather than uniform contribution.

Relation attention determines the final power-state compromise. Its removal slightly improves mean ON MAE and SAE, but lowers macro-F1 from 0.796 to 0.790. It also increases the across-seed standard deviation from 5.35 to 14.69 W for ON MAE and from 0.008 to 0.015 for macro-F1. Thus, cross-appliance exchange does not minimize every power error; it is retained because it improves joint detection and makes the activity-sensitive results more consistent across training runs. The complete model consequently provides the best MAE and macro-F1 rather than the best value for every individual metric.

D. Effects of the Training Strategies

Table VI evaluates focal-event mixing and appliance-wise balancing independently and jointly with the architecture fixed.

TABLE III

POWER AND ON-OFF RESULTS (MEAN ± STANDARD DEVIATION OVER THREE SEEDS; MAE, ON MAE, AND SAE ARE IN W). BOLD DENOTES THE BEST DISPLAYED MEAN, INCLUDING TIES.

Metric	Method	UK-DALE						REFIT					
		Kettle	Fridge	Dishwasher	Washer	Microwave	Avg.	Kettle	Fridge	Dishwasher	Washer	Microwave	Avg.
MAE	UNet-NILM	9.2±2.1	29.1±0.9	22.6±5.1	9.1±0.1	6.1±0.3	15.2±1.1	8.6±0.9	35.0±1.7	12.1±1.1	7.6±0.1	7.4±0.3	14.2±0.3
	MAT-Conv	5.5±1.6	**18.4±0.7**	8.9±2.1	9.6±1.4	5.3±0.4	9.5±1.1	**5.4±0.4**	40.8±0.1	5.4±0.8	6.2±2.3	**6.0±0.7**	12.8±0.6
	Sun *et al.*	8.8±1.7	34.4±3.0	14.7±6.2	15.5±5.0	6.3±0.1	15.9±1.3	10.9±1.6	35.6±1.8	9.0±1.2	18.0±7.3	6.7±0.1	16.1±0.8
	MART-NILM	**3.8±0.2**	19.8±0.7	**4.2±0.4**	**5.3±0.3**	**3.4±0.1**	**7.3±0.2**	5.9±0.1	**33.0±1.2**	**4.9±0.2**	**4.1±0.8**	6.8±0.1	**10.9±0.2**
SAE	UNet-NILM	8.4±2.6	24.5±2.5	21.1±6.5	9.0±0.1	6.0±0.4	13.8±1.3	5.6±1.1	25.9±2.6	9.8±0.3	7.1±0.3	7.0±0.2	11.1±0.7
	MAT-Conv	4.3±1.2	**10.0±1.0**	6.7±1.4	8.8±1.3	5.0±0.4	6.9±0.8	2.9±0.3	23.7±0.6	4.5±1.0	5.0±1.8	**5.2±0.5**	8.3±0.5
	Sun *et al.*	7.3±1.4	32.1±3.5	12.0±5.9	14.5±4.9	6.2±0.1	14.4±1.4	7.1±1.1	21.7±4.4	7.9±1.3	13.0±5.5	6.6±0.0	11.3±0.7
	MART-NILM	**2.9±0.1**	12.8±1.5	**2.6±0.7**	**4.0±0.4**	**2.9±0.1**	**5.1±0.4**	**2.6±0.1**	**15.4±1.6**	**3.7±0.2**	**3.1±0.9**	5.2±0.1	**6.0±0.2**
ON MAE	UNet-NILM	1296.5±443.8	48.4±4.0	747.7±180.9	462.6±7.3	1178.8±93.3	746.8±110.0	1289.1±190.7	63.4±3.4	398.3±93.1	306.1±6.7	1196.7±58.0	650.7±59.1
	MAT-Conv	547.5±180.2	**22.1±1.6**	206.9±46.2	385.7±59.1	946.4±99.8	421.7±75.2	850.7±44.4	46.3±2.5	208.2±21.0	195.6±10.1	1062.2±90.6	472.6±29.3
	Sun *et al.*	807.3±69.1	61.0±7.1	449.8±212.8	478.2±28.0	1215.2±19.2	602.3±44.3	1012.0±65.7	59.1±6.4	437.1±91.3	306.7±17.6	1224.2±5.6	607.8±16.5
	MART-NILM	**383.4±15.9**	25.4±2.1	**87.0±7.4**	**161.8±17.0**	**218.8±23.9**	**175.3±6.9**	**785.2±17.4**	**40.4±1.5**	**139.5±6.0**	**155.9±5.3**	**429.0±40.8**	**310.0±3.5**
AP	UNet-NILM	0.908±0.016	0.873±0.041	0.580±0.274	0.866±0.026	0.535±0.024	0.753±0.049	0.653±0.036	0.657±0.030	0.269±0.056	0.654±0.129	0.136±0.046	0.474±0.024
	MAT-Conv	0.968±0.020	0.951±0.006	0.868±0.039	0.555±0.191	**0.718±0.063**	0.812±0.051	0.801±0.018	0.537±0.005	0.736±0.040	0.653±0.066	0.152±0.046	0.576±0.024
	Sun *et al.*	0.930±0.016	0.915±0.030	0.838±0.094	0.207±0.128	0.514±0.194	0.681±0.075	0.660±0.090	0.641±0.053	0.615±0.040	0.123±0.019	0.160±0.069	0.440±0.000
	MART-NILM	**0.997±0.000**	**0.970±0.001**	**0.959±0.002**	**0.985±0.007**	**0.722±0.046**	**0.926±0.010**	**0.848±0.007**	**0.736±0.030**	**0.759±0.014**	**0.914±0.010**	**0.547±0.056**	**0.761±0.016**
Precision	UNet-NILM	0.80±0.09	0.74±0.09	0.55±0.19	0.84±0.09	0.57±0.07	0.70±0.03	0.63±0.08	0.60±0.12	0.26±0.06	**0.89±0.05**	0.18±0.07	0.51±0.04
	MAT-Conv	0.82±0.10	0.76±0.03	0.47±0.05	0.61±0.18	**0.71±0.09**	0.67±0.08	0.70±0.10	0.59±0.00	0.42±0.06	0.59±0.07	0.16±0.03	0.49±0.04
	Sun *et al.*	0.74±0.08	0.87±0.01	0.51±0.16	0.22±0.09	0.55±0.49	0.58±0.11	0.57±0.08	0.63±0.00	0.54±0.09	0.14±0.02	0.37±0.28	0.45±0.04
	MART-NILM	**0.93±0.01**	**0.89±0.02**	**0.65±0.03**	**0.92±0.01**	**0.72±0.00**	**0.82±0.00**	**0.74±0.01**	**0.64±0.01**	**0.63±0.02**	0.83±0.07	**0.46±0.02**	**0.66±0.01**
Recall	UNet-NILM	0.87±0.08	0.82±0.14	0.65±0.21	0.78±0.04	0.57±0.07	0.74±0.05	0.73±0.03	0.65±0.18	0.46±0.07	0.43±0.13	0.37±0.04	0.53±0.07
	MAT-Conv	0.97±0.01	**0.98±0.02**	0.98±0.02	0.49±0.18	0.67±0.13	0.82±0.02	0.78±0.03	**0.93±0.06**	**0.83±0.02**	0.76±0.21	0.42±0.25	0.74±0.11
	Sun *et al.*	0.92±0.03	0.94±0.01	0.89±0.14	0.56±0.06	0.08±0.07	0.68±0.05	0.75±0.05	0.78±0.01	0.77±0.08	0.50±0.12	0.03±0.02	0.57±0.01
	MART-NILM	**1.00±0.00**	0.94±0.01	**0.99±0.01**	**0.97±0.01**	**0.90±0.02**	**0.96±0.00**	**0.82±0.01**	0.78±0.04	0.75±0.03	**0.88±0.03**	**0.72±0.04**	**0.79±0.02**
F1	UNet-NILM	0.83±0.03	0.77±0.07	0.60±0.20	0.81±0.03	0.57±0.04	0.71±0.03	0.67±0.03	0.61±0.08	0.33±0.07	0.58±0.12	0.23±0.05	0.48±0.04
	MAT-Conv	0.89±0.06	0.86±0.01	0.63±0.04	0.54±0.17	0.68±0.06	0.72±0.05	0.73±0.05	**0.72±0.02**	0.55±0.05	0.65±0.06	0.23±0.08	0.58±0.02
	Sun *et al.*	0.82±0.04	**0.91±0.00**	0.64±0.14	0.30±0.09	0.15±0.13	0.56±0.05	0.64±0.03	0.70±0.00	0.63±0.05	0.21±0.01	0.06±0.04	0.45±0.00
	MART-NILM	**0.96±0.00**	**0.91±0.00**	**0.79±0.02**	**0.94±0.01**	**0.80±0.01**	**0.88±0.00**	**0.78±0.01**	0.71±0.01	**0.69±0.00**	**0.85±0.03**	**0.56±0.03**	**0.72±0.00**

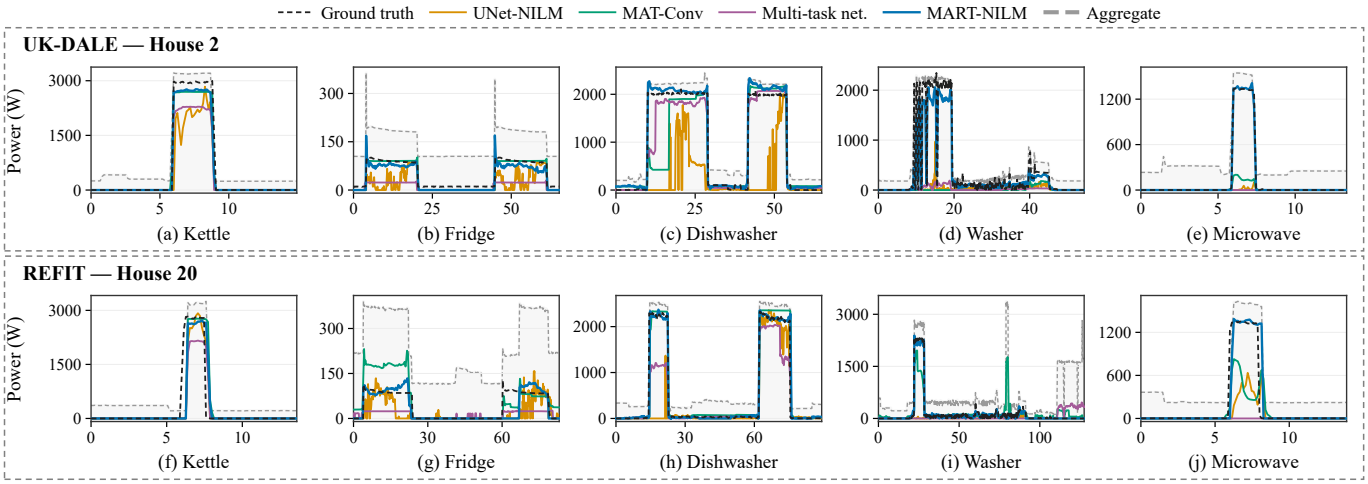


Fig. 5. Waveform comparisons on the two test houses.

TABLE IV

MODEL SIZE AND OBSERVED COMPUTATIONAL COST. M AND MB DENOTE MILLIONS OF PARAMETERS AND MEGABYTES, RESPECTIVELY.

Model	Params (M)	Size (MB)	Training (s)	Inference (s)
UNet-NILM	5.832	22.299	17,455.8	80.40
MAT-Conv	3.022	11.611	10,354.9	14.67
Sun *et al.*	7.829	59.793	16,493.8	23.50
MART-NILM	**1.373**	**5.335**	**485.2**	**1.14**

Focal-event mixing raises macro-F1 by 0.044 and lowers ON MAE by 56.19 W but increases MAE by 0.52 W, indicating an event-coverage rather than general regularization effect. Appliance-wise balancing improves all four measures when used alone. Together, the strategies give the best MAE, SAE, and macro-F1, whereas focal-event mixing alone retains the lowest ON MAE.

TABLE V

ARCHITECTURE ABLATION (THREE-SEED MEAN ± STANDARD DEVIATION OVER THE TWO TEST HOUSES; MAE, ON MAE, AND SAE ARE IN W).

Variant	Params (M)	MAE ↓	ON MAE ↓	SAE ↓	Macro-F1 ↑
Single-scale CNN	1.372	9.22 ±0.18	241.75 ±8.76	5.66 ±0.17	0.795 ±0.001
Without feature encoder	1.326	9.16 ±0.06	252.01 ±2.21	5.59 ±0.07	0.790 ±0.006
Without shared temporal encoder	0.634	10.56 ±0.33	288.07 ±10.25	6.56 ±0.16	0.710 ±0.013
Without appliance decoders	0.836	9.22 ±0.66	265.84 ±4.30	5.63 ±0.63	0.783 ±0.015
Without cross-appliance relation attention	1.332	9.18 ±0.09	**230.84 ±14.69**	**5.55 ±0.04**	0.790 ±0.015
MART-NILM	1.373	**9.12 ±0.14**	242.94 ±5.35	5.58 ±0.17	**0.796 ±0.008**

Figure 6 stratifies performance by background demand, estimated as aggregate power minus the five target powers. Focal-event mixing most clearly improves kettle and microwave reconstruction, consistent with greater exposure to sparse events;

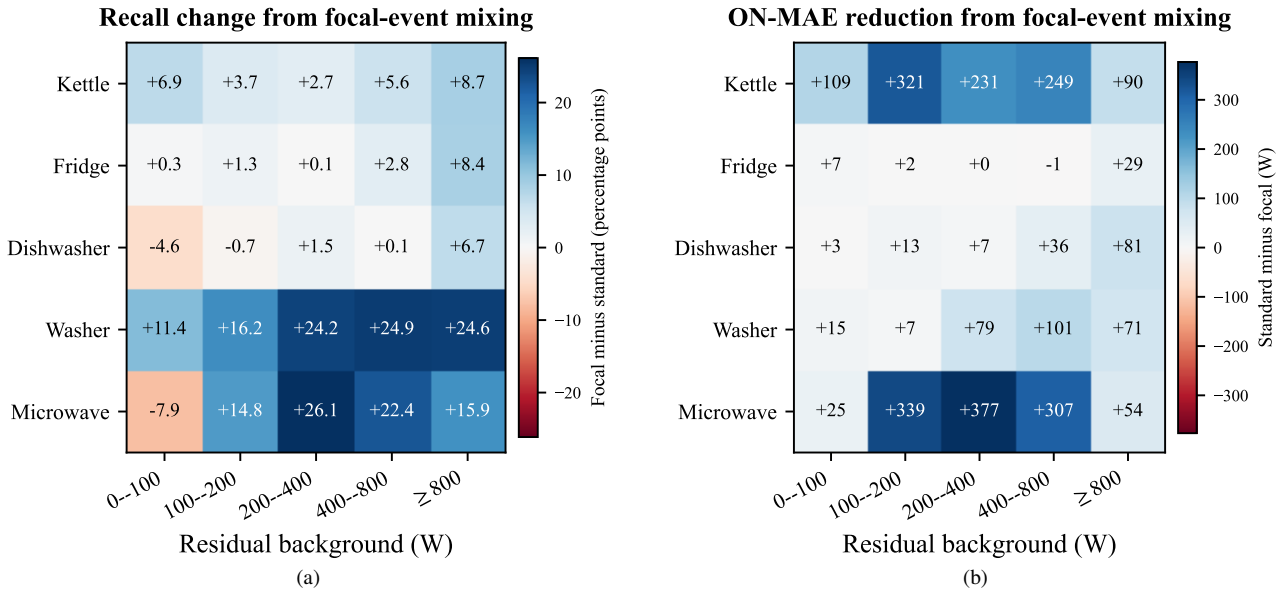


Fig. 6. Effect of focal-event mixing across validation background-demand intervals: (a) active-state recall change in percentage points; (b) ON MAE reduction in watts. Positive values indicate improvement.

TABLE VI
TRAINING-STRATEGY ABLATION ON THE VALIDATION HOUSES (THREE-SEED MEAN ± STANDARD DEVIATION; MAE, ON MAE, AND SAE ARE IN W). † DENOTES A STRATEGY PROPOSED IN THIS WORK.

Training strategy	MAE ↓	ON MAE ↓	SAE ↓	Macro-F1 ↑
Standard joint training	21.164 ±0.394	357.808 ±19.280	15.938 ±0.348	0.605 ±0.012
+ Focal-event mixing†	21.679 ±0.144	**301.623 ±16.196**	15.601 ±0.365	0.649 ±0.016
+ Appliance-wise balancing†	20.929 ±0.065	334.862 ±10.888	15.175 ±0.322	0.661 ±0.014
Complete MART-NILM†	**20.791 ±0.175**	315.408 ±10.405	**15.016 ±0.385**	**0.687 ±0.003**

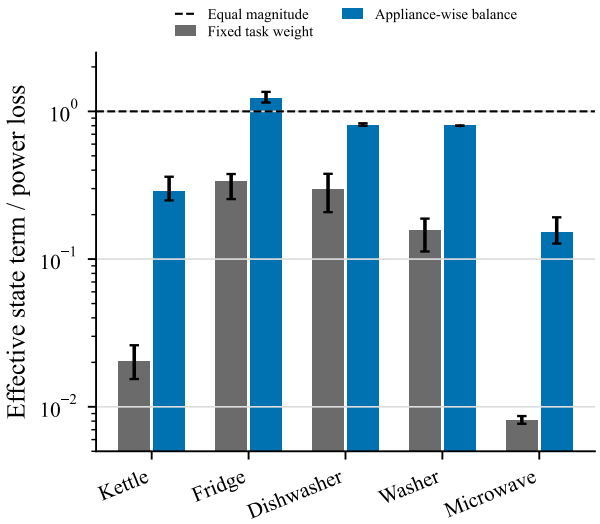


Fig. 7. Power-state scale alignment produced by appliance-wise balancing. Bars show the median effective state-term-to-power-loss ratio pooled over epochs 20 onward and three seeds; error bars denote the interquartile range.

washer recall also improves across most intervals. Some low-background cases deteriorate, so the evidence supports broader coverage rather than uniform gains.

Figure 7 addresses the complementary problem of task-scale mismatch. Under a fixed weight, the median effective state term is only 2.0% and 0.8% of the corresponding power loss for kettle and microwave; balancing raises these ratios to 28.5% and 15.1% and brings the other appliances closer to the power-loss scale. This numerical alignment is consistent with the broader validation gains in Table VI, although it represents loss-scale rather than gradient-norm balancing.

E. Limitations and Future Work

The remaining limitations concern microwave detection and sensitivity to the background-demand level. In the training-strategy ablation, microwave F1 changes from 0.352 ± 0.026 to 0.348 ± 0.027, so no detection gain is claimed. Earlier neural and multi-appliance studies likewise report weaker or highly variable microwave performance [5], [8], [15], while SAMNet shows that similar-power background loads can be mistaken for the target appliance [13]. Short, intermittent, and variable-power activations compound this ambiguity. Focal-event mixing mitigates microwave errors across several background-demand conditions in Fig. 6 but does not eliminate them. Under high aggregate demand, overlapping non-target activity can obscure appliance transitions even when the target signature is unchanged. Domain variation in NILM should therefore encompass both appliance-signature shifts and changes in household background-demand composition. Future work should model these sources jointly across houses and regions rather than treating appliance waveform variation alone.

VI. CONCLUSION

This paper proposed MART-NILM and two complementary training strategies for joint appliance power and state estimation from low-frequency aggregate measurements. The architecture extracts shared temporal evidence, separates appliance-

specific representations, and exchanges relational context before state-gated power prediction. To support learning from sparse activations, focal-event background-stratified mixing varies the background demand surrounding appliance events, while bounded appliance-wise balancing coordinates their power and state objectives. Across three seeds, MART-NILM achieves appliance-average MAEs of 7.3 and 10.9 W and macro-F1 scores of 0.88 and 0.72 on the UK-DALE and REFIT test houses, respectively, outperforming three reproduced baselines on both measures with 1.373 million parameters. The ablations support the complementary roles of temporal modeling, appliance specialization, and the training strategies. Together, these findings support compact joint power-state estimation as a smart-meter observation layer for residential demand-side monitoring.

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